

OOP (Object Oriented Programming)

OOP ka concept ha kisi particular object ke sabhi characterstics ko ek similar object me mimic krna.

Example: Student data:

```
String name = "Aditya";
Int age = 23;
Int rollnumber = 10;
String college = " GDC, Solan"
```

Yha pr “Student” ek real world Object ha, or ye sabhi characterstics isi real world objects ke ander hi ha. It means ki sabhi variables jo alag-alag ha vo Student block ke ander hi mimic hoge OOP me.

Student	
String name	Int rollnumber
Int age	String college

Real World	Programming
Students = ye ek ABSTRACT TERM ha.(ism bhot objects aate ha→ Tanuj, Rahul , etc “or in sabke alg-alg characteristics ha.	In ABSTRACT TERMS ko hum (class students) se represent krta hu. Therefore, class is known as blueprint of object
	Ye class hamesha public class ke bahar hi banegi. or
	Isme jo bhi variables save hote ha unhe INSTANCE VARIABLES khete ha.
	Or isme jo bhi functions save hote ha unhe INSTANCE FUNCTIONS khte ha.

```
public class OOPintroduction {
    public static void main(String[] args) {

        Student s1 = new Student();//
        Student s2 = new Student();

        s1.name = "Rahul";
        s1.age = 18;
        s1.rollnumber = 100;
        s1.college = "G.D.C.Solan";

        s2.name = "Ansh";
        s2.age = 19;
        s2.rollnumber = 101;
        s2.college = "G.D.C.Solan";

        s1.markAttendance();
        s1.print();
        s2.markAttendance();
        s2.print();
    }
}
```

```

    }
    // ha pr public class khatam ha.

class Student{//class is the blueprint of the data.
    String name; // these all are the characterstics of data / object.
    int age;    // these are known as INSTANCE VARIABLES.
    int rollnumber;
    String college;

    // these objects are known as INSTANCE METHODS.
    void markAttendance() { //these all are the behaviour of the data / object.
        System.out.println ("Attendance marked by " + name);
    }
    void print(){
        System.out.println("Name = " + name +", "+ "Age = "+ age+", "+ "Roll Number = "+ rollnumber+", " + "College
Name = "+ college);
    }
}
}

```

Cunstroctors in OOP

Defination: jab bhi object create ho ra hota ha to CUNSTROCTORS use value assign kr dete ha.

Rules of Cunstroctors:

1. Same name as class.
2. It is also a method / function of special type→ With no return; type, and not even “ void”.
3. Automatically called during object creation.
4. Used to initialize an object.
5. It can also be overloaded.

```

class Student { // public class ke bahar bani ha.
    String name ;
    int age;
    int rollNumber;
    String college;

// constructor method / functions below.
    Student(){ // same name as class. Taki details khali hone pr cunstrocter ki details
print ho.
        name = "Priyansh";
        age = 19;
        rollNumber = 1;
        college = "G.D.C Solan";
    }
}

```

Parameterized Cunstroctor;

```

// Parameterized Constructor: it simply means ki hum parameters de rhe ha, uske hisab se
details print kro.
Student(String n, int a, int rN, String c){
    name = n;
    age = a;
    rollNumber = rN;
    college = c;

} // or ab hume khali method call nahi kr skte kuki ab hume parameters assign krne hi
pdege.

```

```
// jab tak hum default parameter add krke constructor overloading na kr le.
```

“This” Keyword

```
class Student { // public class ke bahar bani ha.
    String name ; // niche wala "this" isko denote kr rha ha.
        int age;
        int rollNumber;
        String college;

    Student(String name, int age, int rollNumber, String college){
        this.name = name; // "this." is used to denote class instance variables
        this.age = age;
        this.rollNumber = rollNumber;
        this.college = college;
    }
}
```

Simply “this” student class ke instance variables ki values ko denote krne ke liya ha tki hum same values rkhske.

Overloading in Constructor

```
class Student{
    String name;
    int age;
    int rollNumber;
    String college;

    Student(){}

    Student(String name){
        this.name = name;
    }
    Student(String name, int age){
        this.name = name;
        this.age = age;
    }
    Student(String name, int age, int rollNumber){
        this.name = name;
        this.age = age;
        this.rollNumber = rollNumber;
    }
    Student(String name, int age, int rollNumber, String college){
        this.name = name;
        this.age = age;
        this.rollNumber = rollNumber;
        this.college = college;
    }
}
```